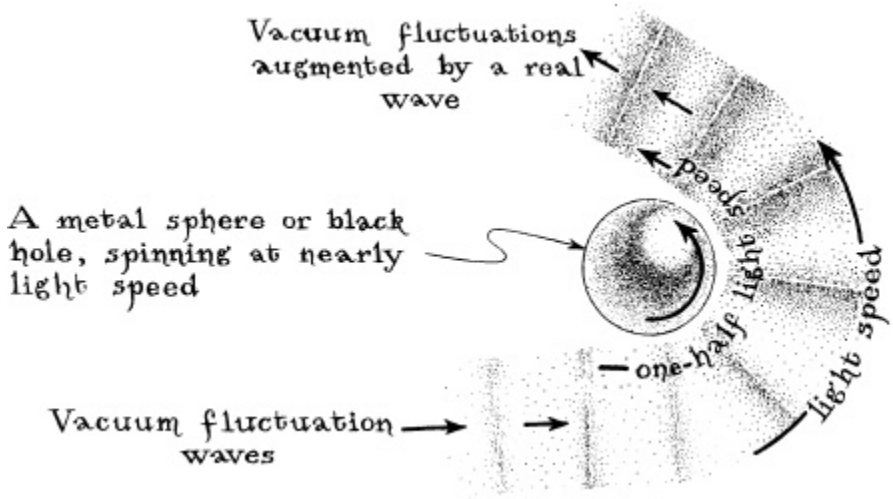
This emission is called *spontaneous* because, when it was first identified as a physical effect, physicists did not realize it was being triggered by vacuum fluctuations. As another example, inside a laser, random electromagnetic vacuum fluctuations interfere with the coherent laser light (interference in the sense of Box 10.3), thereby modulating the laser light in unpredictable ways. This causes the photons emerging from the laser to come out at random, unpredictable times, instead of uniformly one after another—a phenomenon called *photon shot noise.*

Gravitational vacuum fluctuations, by contrast with electromagnetic, have never yet been seen experimentally. Technology of the 1990s, with great effort, should be able to detect highly energetic gravitational waves from black-hole collisions (Chapter 10), but not the waves' far weaker vacuum fluctuations.

---

*This "primary" photon gets absorbed by a phosphor coating on the tube's walls, which in turn emits "secondary" photons that we see as light.



12.1 Zel'dovich's mechanism by which vacuum fluctuations cause a spinning body to radiate.

The incoming wave is not a "real" wave, Zel'dovich explained, but rather a vacuum fluctuation. As this fluctuational wave sweeps around the spinning body, it behaves like a line of ice skaters making a turn: The outer skaters must whip around at high speed while the inner ones move much more slowly; similarly, the wave's outer parts move at a very high speed, the speed of light, while its inner parts move much more slowly than light and, in fact, more slowly than the body's surface is spinning.[8] In such a